

Festigkeitsnachweis - Strength Verification Report

Input Parameters

Parameter	Value	Unit
Material	41Cr4 (EN 10083)	
R _{m,N}	1000.0	MPa
R _{e,N}	800.0	MPa
R _{p,max}	1050.0	MPa
Diameter d	42.0	mm
d/D	0.840	-
σ _a	150.0	MPa
σ _m	0.0	MPa
τ _a	100.0	MPa

Results

Check	Applied	Allowable	Safety Factor
Bending	150.0	1014.7	6.76
Torsion	100.0	1014.7	10.15
Fatigue	229.1	505.6	2.21

Overall Result: PASS